

AI/ML Projects Portfolio

Aditi Garg

Delivery Delay Prediction

Developed a machine learning model to predict delivery delays using historical shipment and logistics datasets. The project involved data preprocessing, feature engineering, exploratory data analysis, and predictive modeling. Implemented algorithms to identify patterns affecting delivery timelines such as traffic, weather conditions, warehouse delays, and route efficiency. Evaluated multiple machine learning models and optimized prediction accuracy.

Skills & Tools Used:

Python, Pandas, NumPy, Scikit-learn, Machine Learning, Data Preprocessing, Feature Engineering, Predictive Analytics, Jupyter Notebook

Voice AI Agent

Built an AI-powered conversational voice assistant capable of handling user queries through natural language interactions. Integrated speech-to-text and text-to-speech workflows with LLM-based conversational intelligence. Designed prompt engineering workflows and API integration for real-time responses. The project focused on improving conversational accuracy and automation efficiency.

Skills & Tools Used:

Python, NLP, LLMs, Prompt Engineering, Speech-to-Text, Text-to-Speech, APIs, Generative AI, LangChain

NCERT Science Study Assistant LLM

Developed an educational AI assistant using Retrieval-Augmented Generation (RAG) to provide contextual answers from NCERT science content. Implemented semantic search and intelligent retrieval mechanisms to improve response quality and concept understanding. Designed prompts for simplified student-friendly explanations and interactive learning.

Skills & Tools Used:

Python, RAG, LLMs, NLP, LangChain, Semantic Search, Prompt Engineering, Generative AI

AdmitGuard - Data Validation System

Worked on a data validation and verification project focused on maintaining data consistency and quality across applications. Implemented validation rules, automated checks, and structured workflows to identify incorrect or missing data. Optimized validation pipelines and improved reporting mechanisms to ensure reliable and accurate datasets.

Skills & Tools Used:

Python, SQL, Data Validation, Automation, Pandas, Backend Development, Reporting, Workflow Optimization

Loan Default Prediction

Built a machine learning solution to predict the probability of loan defaults using customer financial and transactional data. Performed data cleaning, feature engineering, and model training using classification algorithms. Evaluated models using performance metrics and optimized prediction accuracy for financial risk assessment.

Skills & Tools Used:

Python, Machine Learning, Scikit-learn, Classification Models, Data Analysis, Predictive Modeling, Pandas, NumPy